

WASP-10b

R-1638

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_210831 · created 2026-08-02T06:34:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459457.7664 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.06
Transit depth (Rp/Rs)^2	4.42 +/- 2.54 [%]
Orbital Inclination (inc)	86.24 +/- 1.85
Airmass coefficient 1 (a1)	0.53 +/- 0.29
Airmass coefficient 2 (a2)	0.52 +/- 0.71
Scatter in the residuals of the lightcurve fit is	47.94 %
Best Comparison Star	None
Optimal Aperture	7.67
Optimal Annulus	12.15
Transit Duration (day)	0.069 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.21 $\pm$ 0.06 vs 0.1592 (deviation 0.81 σ)
Duration fit vs published	0.069 d vs 0.0946 d (deviation 0.82 σ)
Mid-time O–C	-0.1 min
Depth SNR	3.3
Residual scatter	47.94 %

Light curve

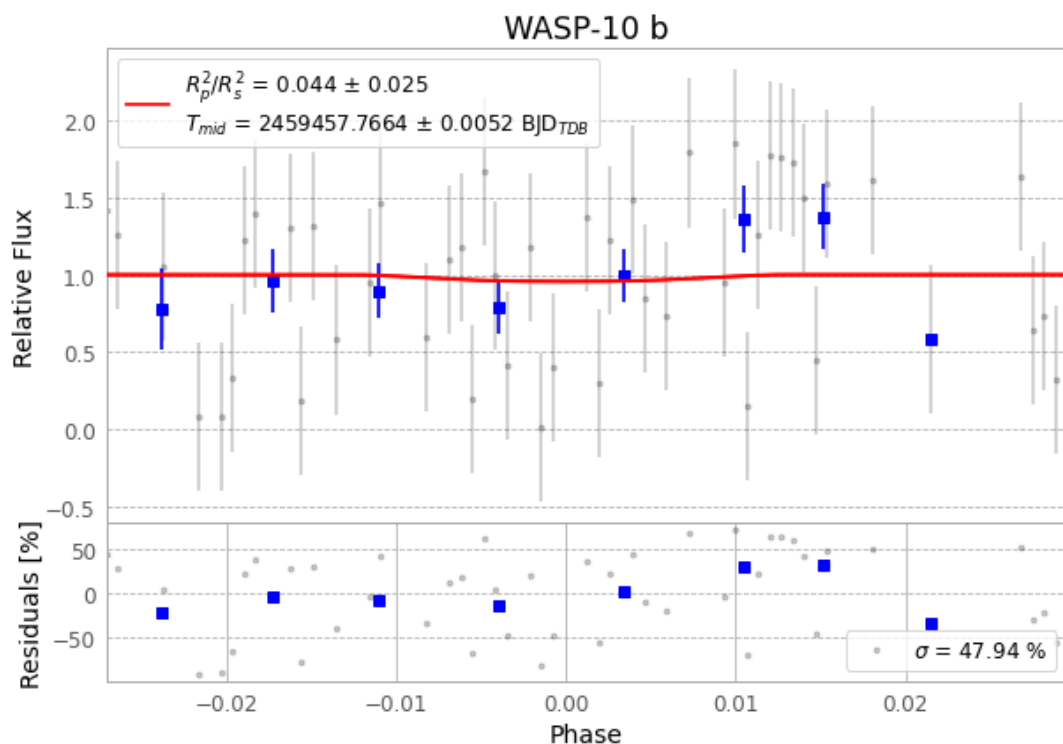


Figure 1 — FinalLightCurve_WASP-10 b_2021-08-30.png (SHA-256 43b361d555c18fd4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 397], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e869264c9b449db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), WASP-10210831041512.FITS → WASP-10210831082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-210831.log (7b6d1e4c0def...), FinalLightCurve_WASP-10 b_2021-08-30.png (43b361d555c1...), FinalLightCurve_WASP-10 b_2021-08-30.csv (36a2c48ebc99...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:34Z.